

Footwear Flux Theory of Geopolitical Deterioration

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Abstract

This paper develops the Footwear Flux Theory of Geopolitical Deterioration. The central proposition is that the geopolitical information contained in footwear is not primarily its absolute level of production or trade, nor the spectacular “shoe incident” as a symbolic event, but the changing rate of economically important footwear flows through an international network. Building upon a sources–stocks–flows–sinks representation of the world economy, the paper distinguishes footwear stocks, footwear flows, footwear flux, network concentration, adaptive capacity, diplomatic stress, symbolic rupture, and militarization. A Footwear Geopolitical Stability Condition is derived from a dynamical-systems representation.

The resulting framework combines international economics, network theory, finance, econometrics, physics, chemistry, biology, psychology, philosophy, religion, political science, diplomacy, international relations, welfare economics, warfare economics, statistics, computer science, and artificial intelligence. The theory is explicitly presented as a falsifiable empirical conjecture rather than an established causal law. The proposed empirical program uses bilateral trade data, diplomatic-event data, and dyadic armed-conflict data to test whether persistent deterioration in economically important footwear relationships precedes diplomatic deterioration and, at longer horizons, militarized conflict.

The paper ends with “The End”

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1 Introduction

The shoe is an unusually rich economic and political object. It is simultaneously a consumer good, a manufactured product, a trade good, a labor-intensive industrial output, a networked supply-chain component, a financial asset of working capital, a cultural symbol, and a physical object capable of entering political communication.

The central proposition of this paper is deliberately narrower than the statement that “shoes predict wars.” The hypothesis is instead that the *rate of change of footwear relationships* may contain information about changing international economic and political relationships.

The distinction is analogous to the distinction between a stock and a flow. A large bilateral footwear trade relationship may be perfectly stable, while a smaller relationship may be deteriorating at an extraordinary rate. Hence the level of trade and the change in trade contain different information.

The present theory builds directly upon a general representation of the world economy as a dynamic system of sources, stocks, flows, and sinks [4]. That framework represents international trade as a network flow and explicitly incorporates network centrality, resilience, time-series analysis, vector error-correction models, machine learning, control theory, and empirical validation [4].

The theory also draws upon the author's distinction between the financial flow and financial flux of a time-dependent interest rate [1, 2]. The associated fundamental identity is

$$F(i(t))f(i(t)) = 1, \quad (1)$$

where $F(i(t))$ denotes financial flux and $f(i(t))$ denotes financial flow [3]. The present paper generalizes the conceptual distinction between flow and flux to international footwear trade.

2 The World Economy as a Dynamical Network

Let the world economy be represented by a directed weighted network

$$G(t) = (V, E(t), W(t)), \quad (2)$$

where V denotes economic entities, $E(t)$ bilateral relationships, and $W(t)$ the associated weights. Let the economic state vector be

$$x(t) \in \mathbb{R}^n. \quad (3)$$

The general dynamics are

$$\frac{dx(t)}{dt} = f(x(t), u(t), d(t), t), \quad (4)$$

where $u(t)$ represents policy interventions and $d(t)$ external disturbances [4].

The world economy may be decomposed into

$$\text{Sources} \longrightarrow \text{Stocks} \longrightarrow \text{Flows} \longrightarrow \text{Sinks}. \quad (5)$$

This representation is compatible with conservation-style reasoning in economics. A generic value identity is

$$\frac{dV_{\text{total}}}{dt} = V_{\text{created}} - V_{\text{destroyed}} + V_{\text{transformed}}. \quad (6)$$

The framework therefore treats economic relationships as dynamical rather than static.

3 Footwear as an Economic Subsystem

Footwear production depends upon physical capital, labor, human capital, materials, technology, logistics, finance, information, and organizational knowledge.

Let the footwear-capability stock of country i be

$$K_i^F(t) = K_i^P(t) + K_i^L(t) + K_i^C(t) + K_i^R(t), \quad (7)$$

where K_i^P is physical productive capacity, K_i^L labor and human-capital capacity, K_i^C commercial and supplier-network capacity, and K_i^R logistics and distribution capacity.

Its dynamics may be written as

$$\frac{dK_i^F}{dt} = I_i(t) - \delta_i K_i^F(t) + A_i(t), \quad (8)$$

where $I_i(t)$ is investment, δ_i depreciation, and $A_i(t)$ technological or organizational augmentation.

The physical production process additionally depends upon materials chemistry. Modern footwear combines leather, rubber, polymers, foams, textiles, adhesives, metals, and composite materials. Consequently, disruptions in petroleum-derived polymers, rubber, tanning inputs, chemical intermediates,

energy, or transportation can propagate through the footwear network even when diplomatic relations themselves have not changed.

This creates a material-network analogue of the economic network:

$$\text{Resources} \rightarrow \text{Materials} \rightarrow \text{Components} \rightarrow \text{Footwear} \rightarrow \text{Trade}. \quad (9)$$

4 The Footwear Flow

Let

$$T_{ij}(t) \geq 0 \quad (10)$$

denote the value or quantity of footwear exported from country i to country j at time t .

A gravity-style representation is

$$T_{ij}(t) = A [K_i^F(t)]^\alpha [D_j(t)]^\beta e^{-\gamma d_{ij}} e^{\phi X_{ij}(t)}, \quad (11)$$

where D_j represents demand, d_{ij} geographical or generalized economic distance, and X_{ij} bilateral trade-facilitation conditions.

This is a footwear-specific specialization of the gravity formulation in the world-economy framework [4].

Contemporary data make the industry especially suitable for network analysis. The World Footwear Yearbook reports that global footwear production reached approximately 23.9 billion pairs in 2024, with Asia accounting for about 88 percent of production. China, India, and Vietnam were the three largest producing countries [7]. In 2024, global footwear exports reached approximately 14.8 billion pairs; China accounted for 62.2 percent of global exports, Vietnam 10.7 percent, and Indonesia 4.1 percent [8].

The 2026 World Footwear Yearbook reports approximately 24.6 billion pairs of production in 2025, with Asia accounting for 88.7 percent [9]. These figures demonstrate that footwear constitutes a highly concentrated global production and trade network.

5 Footwear Flux

Definition 1 (Footwear Flux). *The bilateral footwear flux from country i to country j is the temporal derivative of the bilateral footwear flow:*

$$\Phi_{ij}(t) = \frac{dT_{ij}(t)}{dt}. \quad (12)$$

The normalized footwear flux is

$$\phi_{ij}(t) = \frac{1}{T_{ij}(t)} \frac{dT_{ij}(t)}{dt} = \frac{d \ln T_{ij}(t)}{dt}. \quad (13)$$

The distinction is fundamental:

$$T_{ij}(t) \gg 0 \not\Rightarrow \phi_{ij}(t) \geq 0. \quad (14)$$

A large trade relationship can therefore be contracting rapidly.

Define normalized footwear stress by

$$S_{ij}(t) = -\phi_{ij}(t). \quad (15)$$

Then

$$S_{ij}(t) > 0 \quad (16)$$

indicates contraction of the bilateral footwear flow.

6 Finance and the Flow–Flux Analogy

The author’s earlier treatment of a time-dependent interest rate distinguishes financial flow from financial flux [1, 2]. The corresponding expressions are

$$F(i(t)) = i(t) \frac{(1 + i(t))^t}{(1 + i(t))^t - 1}, \quad (17)$$

$$f(i(t)) = \frac{(1 + i(t))^t - 1}{i(t)(1 + i(t))^t}. \quad (18)$$

Their product is

$$F(i(t))f(i(t)) = 1. \quad (19)$$

The present theory does not assert that international trade satisfies this identity. Rather, the earlier financial framework motivates the methodological distinction between a level of flow and a transformation of that flow.

The analogy is:

$$\text{financial flow} \longleftrightarrow \text{economic trade flow}, \quad (20)$$

and

$$\text{financial flux} \longleftrightarrow \text{rate of change of trade flow}. \quad (21)$$

The footwear hypothesis therefore concerns the *dynamic geometry* of international trade rather than merely its static magnitude.

7 Network Dependence and Concentration

Define bilateral footwear dependence of importer j upon exporter i as

$$D_{ij}(t) = \frac{T_{ij}(t)}{\sum_k T_{kj}(t)}. \quad (22)$$

Define the footwear supplier concentration index

$$H_j(t) = \sum_i D_{ij}^2(t). \quad (23)$$

A high value of H_j indicates concentrated sourcing.

Define geopolitical footwear exposure by

$$G_{ij}(t) = D_{ij}(t)H_j(t)S_{ij}(t). \quad (24)$$

The same footwear shock can therefore have different consequences depending upon the availability of alternative suppliers.

8 Adaptive Capacity

A biological analogy is useful here. Biological systems do not respond to perturbations solely according to the magnitude of the disturbance; resilience depends upon redundancy, adaptation, resource availability, and network architecture.

Economic networks possess analogous properties.

Define bilateral adaptive capacity as

$$A_{ij}(t) = a_1 K_i^F(t) + a_2 K_j^F(t) + a_3 R_{ij}(t) + a_4 N_{ij}(t), \quad (25)$$

where R_{ij} represents relational resilience and N_{ij} the availability of alternative suppliers, markets, transport routes, and institutional arrangements.

Define net footwear stress as

$$\Psi_{ij}(t) = S_{ij}(t) - A_{ij}(t). \quad (26)$$

Thus

$$\Psi_{ij}(t) \leq 0 \quad \text{indicates absorptive capacity,} \quad (27)$$

$$\Psi_{ij}(t) > 0 \quad \text{indicates excess stress.} \quad (28)$$

9 Footwear Geopolitical Stability Condition

Conjecture 1 (Footwear Geopolitical Stability Condition). *A bilateral footwear relationship is locally stable with respect to footwear-sector disturbances when adaptive capacity is at least as large as the normalized rate of footwear-flow deterioration:*

$$A_{ij}(t) \geq -\frac{\dot{T}_{ij}(t)}{T_{ij}(t)}. \quad (29)$$

Equivalently,

$$\Psi_{ij}(t) \leq 0. \quad (30)$$

When

$$\Psi_{ij}(t) > 0, \quad (31)$$

the relationship enters a footwear-induced instability regime.

Persistence is incorporated through cumulative stress:

$$C_{ij}(t) = \int_{t_0}^t \max\{\Psi_{ij}(\tau), 0\} d\tau. \quad (32)$$

A critical threshold C_{ij}^* produces the regime classification

$$C_{ij}(t) = 0, \quad \text{stable,} \quad (33)$$

$$0 < C_{ij}(t) \leq C_{ij}^*, \quad \text{absorbing stress,} \quad (34)$$

$$C_{ij}(t) > C_{ij}^*, \quad \text{persistent instability.} \quad (35)$$

10 Network Amplification and Physics

Let the footwear-network adjacency matrix be

$$A_F(t) = [a_{ij}(t)]. \quad (36)$$

Let $s(t)$ be the vector of bilateral footwear stresses. A linearized propagation equation is

$$s(t+1) = A_F(t)s(t) + \epsilon(t). \quad (37)$$

Iterating yields

$$s(t+h) = A_F^h(t)s(t) + \sum_{k=0}^{h-1} A_F^k(t)\epsilon(t+h-1-k). \quad (38)$$

The spectral radius

$$\rho(A_F) = \max_{\ell} |\lambda_{\ell}(A_F)| \quad (39)$$

determines the local amplification properties of the normalized linear system.

Theorem 1 (Linearized Footwear Network Stability). *Consider the homogeneous system*

$$s(t+1) = A_F s(t). \quad (40)$$

If

$$\rho(A_F) < 1, \quad (41)$$

then $s(t) \rightarrow 0$ for every initial state. If

$$\rho(A_F) > 1, \quad (42)$$

there exists an initial state for which the perturbation grows asymptotically.

Proof. By the spectral decomposition of a finite-dimensional matrix, the asymptotic behavior of repeated multiplication by A_F is governed by its eigenvalues. If the spectral radius is less than one, every eigenmode decays geometrically. If the spectral radius exceeds one, an eigenmode associated with an eigenvalue whose modulus exceeds one grows geometrically. The unit-root boundary $\rho(A_F) = 1$ is the critical case. $\square$

This provides the network-theoretic version of the stability logic already developed in the broader world-economy framework [4].

11 Diplomatic Stress

Let

$$R_{ij}(t) \quad (43)$$

denote accumulated diplomatic stress.

A simple state equation is

$$R_{ij}(t) = \lambda R_{ij}(t-1) + \Delta_{ij}(t), \quad 0 < \lambda < 1, \quad (44)$$

where Δ_{ij} represents new diplomatic deterioration.

The central empirical relationship is

$$\Delta_{ij}(t+1) = f(G_{ij}(t), X_{ij}(t)) + \varepsilon_{ij,t}, \quad (45)$$

where X_{ij} denotes political, economic, military, geographical, and institutional controls.

The hypothesis is not that footwear mechanically causes diplomacy to deteriorate. Rather,

$$\frac{\partial E[\Delta_{ij}(t+1) | G_{ij}(t)]}{\partial G_{ij}(t)} > 0 \quad (46)$$

is the empirical prediction.

12 The Shoe Incident as Symbolic Rupture

Let

$$Q_{ij}(t) \in \{0, 1\} \quad (47)$$

denote a documented shoe-related symbolic diplomatic incident.

The theory does not treat Q_{ij} as a primitive cause of war. It is instead a possible observable manifestation of accumulated political and diplomatic stress:

$$P(Q_{ij,t} = 1) = F(R_{ij,t}, Z_{ij,t}). \quad (48)$$

The famous 1960 Khrushchev incident illustrates the distinction. Contemporary U.S. diplomatic historical records report that Khrushchev removed his shoe, waved it at Philippine delegate Lorenzo Sumulong, and struck his desk with it; he later waved the shoe again during remarks by U.S. official Francis O. Wilcox, after which General Assembly President Frederick Boland adjourned the meeting [15]. The United Nations itself describes the event as a historical legend whose precise visual documentation is disputed [16].

Thus the symbolic incident should not be confused with a mechanical predictor of war.

12.1 The George W. Bush Shoe Incident: Symbolic Rupture Within an Existing Conflict

A second historically important footwear incident occurred in Baghdad on 14 December 2008, when Iraqi journalist Muntadhar al-Zaidi threw his shoes toward United States President George W. Bush during a press conference. The incident occurred in the context of the Iraq War, which had begun in 2003, and therefore provides an important qualification to any interpretation of footwear incidents as predictors of war.

The Bush incident should not be interpreted as evidence that a shoe incident caused, or predicted the beginning of, the Iraq War. The temporal ordering is the reverse: the military conflict was already underway when the symbolic footwear incident occurred. The event is therefore better interpreted within the present framework as an observable manifestation of symbolic rupture following a prolonged period of geopolitical, military, diplomatic, and social deterioration.

Let $Q_{ij}(t)$ denote the intensity of observable symbolic rupture between actors i and j . A footwear incident may constitute an observation of $Q_{ij}(t)$, but it is not necessarily a primitive determinant of $Q_{ij}(t)$. A stylized relationship is

$$Q_{ij}(t) = \mathcal{Q}(\Psi_{ij}(t), R_{ij}(t), X_{ij}(t), \varepsilon_{ij}(t)), \quad (49)$$

where $\Psi_{ij}(t)$ denotes accumulated economic and network stress, $R_{ij}(t)$ denotes diplomatic stress, $X_{ij}(t)$ denotes other political and institutional conditions, and $\varepsilon_{ij}(t)$ represents unobserved influences.

The Bush incident consequently illustrates an important distinction between *prediction of deterioration* and *measurement of rupture*. If a shoe incident occurs after war has already begun, it cannot provide out-of-sample evidence that footwear dynamics predicted the war's onset. It can nevertheless provide observational evidence that footwear can become a highly visible symbolic medium through which accumulated political conflict is expressed.

This distinction can be represented by the temporal ordering

$$\text{Economic Stress} \rightarrow \text{Political Stress} \rightarrow \text{Diplomatic Deterioration} \rightarrow \text{Military Conflict} \rightarrow \text{Symbolic Rupture}. \quad (50)$$

The ordering is not universal and should not be imposed as an empirical necessity. In other cases, symbolic rupture may precede military escalation. The empirical task is therefore to estimate the timing and direction of the relationships rather than to assume that every footwear incident occupies the same position in the causal sequence.

The Bush incident is particularly useful because it demonstrates that the same observable phenomenon can have radically different predictive interpretations depending upon its temporal position in the geopolitical system. A footwear incident occurring before measurable deterioration would constitute one type of observation; a footwear incident occurring during diplomatic deterioration would constitute another; and a footwear incident occurring during an established war, as in Baghdad in 2008, constitutes a third.

Accordingly, the theory distinguishes three empirical categories:

$$\mathcal{F} = \{\mathcal{F}_{\text{pre}}, \mathcal{F}_{\text{diplomatic}}, \mathcal{F}_{\text{conflict}}\}, \quad (51)$$

where $\mathcal{F}_{\text{pre}}$ denotes footwear incidents preceding major geopolitical deterioration, $\mathcal{F}_{\text{diplomatic}}$ denotes incidents occurring during substantial diplomatic deterioration, and $\mathcal{F}_{\text{conflict}}$ denotes incidents occurring after military conflict has already begun.

The George W. Bush incident belongs to the third category. Its principal value for the present theory is consequently not as a leading indicator of war, but as a case of *symbolic rupture embedded within an existing conflict system*. This prevents the theory from committing the common post hoc error of treating a visually dramatic event as evidence for a causal mechanism that preceded it.

The broader implication is methodological. The footwear variable must be timestamped relative to the geopolitical event being predicted. A model that uses $Q_{ij}(t)$ to forecast conflict at $t+h$ must ensure that the footwear observation occurs strictly before the forecast origin and that events occurring after conflict onset are not inadvertently incorporated into the training sample as predictors of conflict onset.

Thus, the Bush incident is not a counterexample to the theory. It is a reason to make the theory more precise. It demonstrates why the distinction between *flow*, *flux*, *stress*, and *symbolic rupture* must be maintained:

$$\text{Footwear Flow} \neq \text{Footwear Flux} \neq \text{Geopolitical Stress} \neq \text{Symbolic Rupture}. \quad (52)$$

The incident therefore belongs in the empirical architecture of the theory as an observed symbolic endpoint or intermediate manifestation, rather than as an assumed causal trigger of war.

13 Philosophy and Semiotics

A shoe is a material object with multiple simultaneous meanings.

In ordinary economics it possesses use value and exchange value. In semiotics it can acquire symbolic value. In phenomenology it belongs to the embodied human world. In political theory, the introduction of an ordinary bodily object into a highly formal institution can function as anti-ritual communication.

The institutional diplomatic sequence is approximately

$$\text{Representative} \rightarrow \text{Speech} \rightarrow \text{Procedure} \rightarrow \text{Institution}. \quad (53)$$

A shoe incident introduces

$$\text{Body} \rightarrow \text{Object} \rightarrow \text{Gesture} \rightarrow \text{Symbol}. \quad (54)$$

The resulting phenomenon is an instance of embodied political communication.

The distinction between object and symbol is crucial:

$$\text{Physical object} + \text{cultural code} = \text{political meaning}. \quad (55)$$

The shoe therefore illustrates the philosophical proposition that political institutions depend not only upon formal rules but upon embodied compliance with institutional rituals.

14 Religion and Sacred Space

Footwear also has a long history of religious symbolism. In Exodus 3:5, Moses is instructed to remove his sandals because the ground is holy [17]. In Qur'an 20:12, Moses is likewise instructed to remove his sandals because he is in the sacred valley of Tuwa [18].

These traditions demonstrate an important semiotic reversal:

$$\text{shoe} \longleftrightarrow \text{boundary between ordinary and sacred space}. \quad (56)$$

The political shoe incident can therefore be interpreted cautiously as an inversion of a broader cultural grammar: footwear can mark the boundary between dignity, contamination, ordinary space, sacred space, and institutional space.

The present theory does not claim that one religious symbolism is universally shared. Cultural interpretation remains historically and geographically contingent.

15 Psychology and Psychiatry

The psychological contribution concerns threat perception, appraisal, affect, identity, humiliation, and escalation rather than psychiatric diagnosis.

Let an actor's perceived threat be

$$\Theta_i(t) = g(E_i(t), I_i(t), P_i(t), H_i(t)), \quad (57)$$

where E_i denotes economic exposure, I_i identity considerations, P_i political incentives, and H_i perceived humiliation or status loss.

Repeated negative signals can alter the perceived state of an opponent even when material changes are initially small.

From a psychological perspective,

$$\text{economic shock} \rightarrow \text{appraisal} \rightarrow \text{attribution} \rightarrow \text{emotion} \rightarrow \text{political response} \quad (58)$$

is a possible escalation pathway.

Psychiatry contributes a caution: symbolic political behavior should not be converted into clinical diagnoses of public actors. The relevant analytical objects are observable behavior, institutional incentives, threat perception, and escalation dynamics.

16 Political Science and International Relations

International relations provides several competing mechanisms.

First, commercial interdependence may raise the opportunity cost of conflict. Recent empirical work using global bilateral data reports a substantial “peace dividend” from increased international trade [14].

Second, asymmetric dependence may create bargaining power.

Third, economic decoupling may reduce the opportunity cost of confrontation.

Fourth, strategic sectors may become instruments of coercion.

Hence the theory does not predict

$$\text{more trade} \rightarrow \text{more conflict.} \quad (59)$$

Instead, the relevant mechanism is potentially

$$\begin{array}{c} \text{important} \\ \text{interdependence} \end{array} + \begin{array}{c} \text{rapid} \\ \text{deterioration} \end{array} + \begin{array}{c} \text{low} \\ \text{substitutability} \end{array} \rightarrow \begin{array}{c} \text{increased} \\ \text{strategic} \\ \text{stress} \end{array}. \quad (60)$$

This distinction is essential because a trade relationship can simultaneously be economically beneficial and strategically vulnerable.

17 Welfare Economics

Let social welfare in country i be

$$W_i = W_i(C_i, L_i, P_i, S_i), \quad (61)$$

where C_i denotes consumption, L_i labor outcomes, P_i production, and S_i strategic security.

Trade may increase consumption and production efficiency while simultaneously increasing exposure to supply disruption.

Thus

$$\frac{\partial W_i}{\partial T_{ij}} > 0 \quad (62)$$

under conventional gains-from-trade assumptions does not imply

$$\frac{\partial W_i}{\partial \text{dependence}} > 0 \quad (63)$$

under all geopolitical conditions.

The welfare optimum may therefore involve diversification rather than maximal bilateral specialization.

18 Warfare Economics

A state contemplating coercion compares expected benefits and costs.

Let the expected net strategic payoff be

$$U_i^W = B_i^W - C_i^W - C_i^E - C_i^P, \quad (64)$$

where B_i^W is expected strategic benefit, C_i^W military cost, C_i^E economic cost, and C_i^P political cost. Footwear dependence enters through C_i^E .

A highly integrated footwear relationship may therefore discourage conflict because disruption is costly. But once alternative supply chains have been established, the same relationship may no longer provide the same deterrent.

This produces a dynamic mechanism:

$$\begin{array}{c} \text{trade} \\ \text{integration} \end{array} \rightarrow \text{dependence} \rightarrow \text{diversification} \rightarrow \begin{array}{c} \text{reduced} \\ \text{dependence} \end{array} \rightarrow \begin{array}{c} \text{lowereconomiccost} \\ \text{ofconfrontation} \end{array}. \quad (65)$$

The theory therefore treats *decoupling* as potentially more geopolitically informative than trade volume itself.

19 Chemistry, Biology, and Industrial Ecology

Footwear is a material system. Chemical inputs are transformed into polymers, foams, adhesives, coatings, dyes, rubber compounds, and other components. The resulting production chain behaves like an industrial ecosystem.

Let material stocks be

$$M(t) = (M_1(t), \dots, M_m(t)). \quad (66)$$

Their dynamics may be written as

$$\frac{dM_k}{dt} = S_k(t) - C_k(t) - D_k(t), \quad (67)$$

where S_k denotes material supply, C_k consumption, and D_k depletion or disposal.

The analogy with biology is network resilience: redundant suppliers resemble ecological redundancy, while highly concentrated supply chains resemble systems with low functional diversity.

This analogy is heuristic rather than an assertion that economies obey biological laws.

20 Econometrics

The primary empirical unit is the country-pair-year observation (i, j, t) .

The bilateral footwear flux estimator is

$$\hat{\phi}_{ij,t} = \ln(T_{ij,t} + c) - \ln(T_{ij,t-1} + c), \quad (68)$$

where $c > 0$ prevents undefined values when bilateral trade is zero.

The primary stress variable is

$$\hat{S}_{ij,t} = -\hat{\phi}_{ij,t}. \quad (69)$$

A fixed-effects diplomatic model is

$$D_{ij,t+1} = \alpha_{ij} + \gamma_t + \beta G_{ij,t} + \theta' X_{ij,t} + \varepsilon_{ij,t}. \quad (70)$$

Here α_{ij} absorbs time-invariant bilateral characteristics and γ_t absorbs global shocks.

The conflict model is

$$P(M_{ij,t+h} = 1) = F(G_{ij,t}, D_{ij,t}, X_{ij,t}), \quad (71)$$

for horizons

$$h \in \{1, 2, 3, 5\}. \quad (72)$$

The stronger prediction is

$$G_{ij,t} \rightarrow D_{ij,t+1} \rightarrow M_{ij,t+h}. \quad (73)$$

The temporal ordering is critical for reducing reverse-causality concerns.

21 Identification and Falsification

The hypothesis should be subjected to deliberately hostile tests.

First, compare footwear with unrelated product categories.

Table 1: Proposed Placebo Tests

Sector	Purpose
Footwear	Primary treatment sector
Textiles	Related labor-intensive sector
Furniture	Consumer manufacturing placebo
Machinery	Capital-intensive placebo
Electronics	Technology-intensive placebo
Food	Essential-goods placebo

If every sector predicts diplomatic deterioration equally well, footwear possesses no special explanatory power.

Second, conduct lead tests:

$$D_{ij,t} = \sum_{k=1}^K \beta_{-k} G_{ij,t-k} + \sum_{k=1}^K \beta_{+k} G_{ij,t+k} + \alpha_{ij} + \gamma_t + \varepsilon_{ij,t}. \quad (74)$$

If diplomatic deterioration systematically precedes footwear stress, rather than follows it, the proposed causal interpretation is weakened.

Third, use alternative definitions of footwear stress, alternative transformations, and alternative conflict outcomes.

22 Data Architecture

The empirical system can combine the following datasets.

Table 2: Core Empirical Data Architecture

Layer	Dataset	Main Use
Footwear trade	CEPII BACI	Bilateral product-level footwear flows
Industry structure	World Footwear Yearbook	Production, consumption, exports, imports
Diplomacy	GDEL	Bilateral political events and event intensity
Armed conflict	UCDP Dyadic Dataset	Dyad-year conflict outcomes
Macroeconomics	World Bank, IMF, OECD	Controls and economic conditions

The current BACI release provides bilateral trade for approximately 200 countries and roughly 5,000 products at six-digit HS classification; its HS92 series covers 1995–2024 [6]. GDEL records bilateral events using the CAMEO taxonomy and supplies event-level fields including the Goldstein scale [10]. UCDP provides a dyad-year Armed Conflict Dataset through 2025 [12].

23 Statistical Testing

The principal null hypothesis is

$$H_0 : \beta = 0. \quad (75)$$

The alternative is

$$H_1 : \beta > 0. \quad (76)$$

For conflict prediction, a stronger hypothesis is

$$H_1^C : \frac{\partial P(M_{ij,t+h} = 1)}{\partial G_{ij,t}} > 0. \quad (77)$$

Granger causality can be examined through

$$D_t = \sum_{k=1}^p a_k D_{t-k} + \sum_{k=1}^p b_k G_{t-k} + \varepsilon_t, \quad (78)$$

with the null

$$H_0 : b_1 = b_2 = \dots = b_p = 0. \quad (79)$$

However, Granger causality is predictive precedence, not structural causality. Identification therefore requires fixed effects, controls, placebo sectors, pre-trend tests, alternative specifications, and ideally quasi-experimental variation.

24 Machine Learning and Early Warning

The problem can also be formulated as a classification task.

Let the feature vector be

$$z_{ij,t} = [T_{ij,t}, \phi_{ij,t}, S_{ij,t}, D_{ij,t}, H_j(t), C_{ij,t}, R_{ij,t}, X_{ij,t}]. \quad (80)$$

A classifier estimates

$$\hat{p}_{ij,t+h} = P(M_{ij,t+h} = 1 \mid z_{ij,t}). \quad (81)$$

Models may include logistic regression, random forests, gradient boosting, recurrent neural networks, temporal transformers, state-space models, and graph neural networks.

A graph neural network is particularly natural because the underlying object is a dynamic graph:

$$G_F(t) = (V, E_F(t), W_F(t)). \quad (82)$$

The machine-learning model must nevertheless be evaluated out of sample. In-sample classification accuracy would provide little evidence of geopolitical forecasting capability.

25 Algorithmic Implementation

The following pseudocode describes the proposed empirical pipeline.

```

INPUT:
bilateral footwear trade T[i,j,t]
diplomatic events D[i,j,t]
conflict observations M[i,j,t]
macroeconomic controls X[i,j,t]

FOR each country pair (i,j):
construct footwear flow T[i,j,t]
calculate normalized footwear flux
phi[i,j,t] = log(T[i,j,t] + c)
- log(T[i,j,t-1] + c)

calculate dependence D[i,j,t]
calculate supplier concentration H[j,t]
calculate footwear stress S[i,j,t] = -phi[i,j,t]

estimate adaptive capacity A[i,j,t]
calculate net stress
Psi[i,j,t] = S[i,j,t] - A[i,j,t]
```

```
calculate geopolitical footwear exposure
 $G[i,j,t] = D[i,j,t] * H[j,t] * \Psi[i,j,t]$ 
```

MERGE:

G with diplomatic and conflict datasets

ESTIMATE:

```
fixed-effects diplomatic regressions
event studies
lagged conflict models
placebo-sector models
Granger-causality models
```

TRAIN:

machine-learning early-warning models

VALIDATE:

```
rolling out-of-sample forecasts
precision
recall
AUROC
calibration
false-alarm rate
```

REPORT:

```
coefficient estimates
confidence intervals
robustness tests
forecasting performance
failure cases
```

26 System Architecture

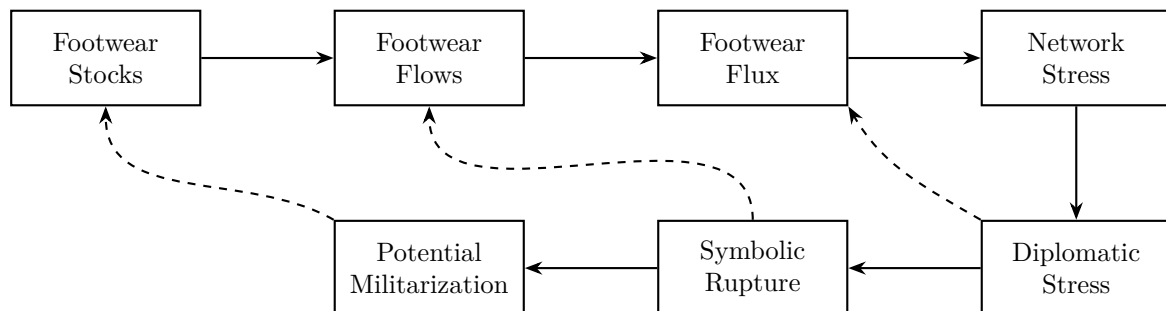


Figure 1: System architecture.

The dashed arrows represent feedback and adaptation. The system is therefore not a one-way causal chain but a coupled dynamical system.

27 Control-Theoretic Interpretation

Let the state vector be

$$x(t) = \begin{bmatrix} K^F(t) \\ T(t) \\ S(t) \\ R(t) \end{bmatrix}. \quad (83)$$

Let policy and external disturbances be $u(t)$ and $d(t)$.
Then

$$\dot{x}(t) = f(x(t), u(t), d(t), t). \quad (84)$$

A linearized system around an equilibrium x^* is

$$\dot{\tilde{x}}(t) = J\tilde{x}(t) + B\tilde{u}(t) + E\tilde{d}(t), \quad (85)$$

where J is the Jacobian.
Local stability requires

$$\text{Re}(\lambda_k(J)) < 0 \quad \text{for all } k. \quad (86)$$

This parallels the stability condition developed in the broader world-economy framework [4].

The economic interpretation is straightforward: a footwear network is resilient when perturbations decay faster than political and economic feedback mechanisms amplify them.

28 Diplomatic Feedback

Suppose diplomatic deterioration itself feeds back into footwear trade:

$$\frac{dT_{ij}}{dt} = F(K_i, K_j, D_j, X_{ij}, R_{ij}). \quad (87)$$

Then

$$\frac{\partial T_{ij}}{\partial R_{ij}} < 0 \quad (88)$$

represents the possibility that worsening diplomacy further contracts trade.
We therefore obtain a feedback loop:

$$\text{Footwear Stress} \rightarrow \text{Diplomatic Stress} \rightarrow \text{Trade Contraction} \rightarrow \text{Greater Footwear Stress}. \quad (89)$$

Such positive feedback is a candidate mechanism for nonlinear escalation.

29 Welfare, Diversification, and Resilience

Suppose diversification has cost $C(D)$ but reduces expected disruption loss $L(D)$.

A planner solves

$$\max_D [W(D) - C(D) - L(D)]. \quad (90)$$

The optimal diversification condition is

$$W'(D) - C'(D) - L'(D) = 0. \quad (91)$$

Thus complete specialization need not be socially optimal when geopolitical disruption is possible.

The theory therefore links trade policy, industrial policy, welfare economics, and national security without assuming that autarky is desirable.

30 International Trade and Decoupling

The current footwear industry illustrates the distinction between production concentration and trade diversification. World Footwear reports that China remains the dominant exporter while Vietnam and Indonesia have substantial and growing positions [8]. The 2025 data similarly show China's continuing dominance alongside persistent Asian concentration [9].

This creates a strategic question:

$$\text{Is diversification} = \text{resilience} \quad (92)$$

or

$$\text{Is diversification} = \text{decoupling}. \quad (93)$$

The answer depends upon the purpose and direction of the reconfiguration.

From a welfare perspective, diversification can reduce expected disruption losses.

From a geopolitical perspective, diversification can simultaneously reduce mutual dependence and therefore reduce the economic cost of confrontation.

This dual effect is central to the theory.

31 The Shoe Incident Reinterpreted

The theory therefore rejects the simplistic chain

$$\text{shoe incident} \rightarrow \text{war}. \quad (94)$$

Instead, the proposed sequence is

$\begin{array}{ccccccc} \textit{Footwear} & \rightarrow & \textit{Footwear} & \rightarrow & \textit{Footwear} & \rightarrow & \textit{Economic} \\ \textit{Stocks} & & \textit{Flows} & & \textit{Flux} & & \textit{Stress} \end{array}$	$\rightarrow$	$\begin{array}{ccccccc} \textit{Diplomatic} & \rightarrow & \textit{Symbolic} & \rightarrow & \textit{Potential} \\ \textit{Stress} & & \textit{Rupture} & & \textit{Military} \end{array}$
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(95)

The shoe incident is therefore a possible *terminal observable*, not necessarily the causal origin of the process.

The 1960 UN episode is instructive precisely because it demonstrates symbolic rupture within a highly formal diplomatic institution, while not itself constituting a general predictor of subsequent interstate war [15, 16].

32 Research Propositions

Conjecture 2 (Footwear Flux Hypothesis). *Persistent negative normalized footwear flux in an economically important bilateral relationship contains incremental information about subsequent diplomatic deterioration after controlling for country-pair effects, global shocks, macroeconomic conditions, and alternative trade shocks.*

Conjecture 3 (Footwear Geopolitical Instability Hypothesis). *The predictive association between footwear stress and diplomatic deterioration increases with bilateral dependence and supplier concentration.*

Conjecture 4 (Network Amplification Hypothesis). *Footwear shocks propagate more strongly through networks with greater centrality and spectral amplification.*

Conjecture 5 (Conflict Extension). *Persistent footwear-network stress contains incremental predictive information about subsequent militarized interstate conflict at horizons longer than the initial trade shock.*

None of these conjectures should be accepted without empirical testing.

33 What Would Falsify the Theory?

The theory would be weakened or rejected if any of the following occur systematically:

1. footwear flux does not predict diplomatic deterioration;
2. diplomatic deterioration consistently precedes footwear deterioration;
3. footwear performs no better than placebo sectors;
4. the result disappears under bilateral fixed effects;
5. the result disappears after controlling for general trade contraction;
6. the apparent relationship is driven entirely by global crises;
7. machine-learning models obtain no out-of-sample predictive improvement from footwear variables;
8. the relationship exists only contemporaneously and not at meaningful predictive horizons.

This falsification structure is essential. A theory that can explain every outcome is not a useful empirical theory.

34 Conclusion

The central contribution of the Footwear Flux Theory is not the claim that shoes cause wars. It is the proposition that an ordinary traded object can be treated as an observable component of a much larger economic and geopolitical dynamical system.

The fundamental distinction is

$$\boxed{\text{Flow is the observable; flux is the signal; network structure is the amplifier.}} \quad (96)$$

A static trade statistic tells us how much footwear moves.

A footwear flux tells us how rapidly that relationship is changing.

A network measure tells us how consequential that change may be.

An adaptive-capacity measure tells us whether the system can absorb it.

Diplomatic data tell us whether political relations subsequently deteriorate.

Conflict data tell us whether the deterioration ultimately crosses into militarization.

The resulting conceptual architecture is therefore

$$\boxed{Sources \rightarrow Stocks \rightarrow Flows \rightarrow Flux \rightarrow \begin{matrix} Network \\ Stress \end{matrix} \rightarrow Diplomacy \rightarrow \begin{matrix} Symbolic \\ Rupture \end{matrix} \rightarrow Conflict.} \quad (97)$$

The most important empirical question is consequently not

$$\text{“Do shoes predict wars?”} \quad (98)$$

but

$$\boxed{\text{“Does the shoe begin to change before the diplomats begin to break?”}} \quad (99)$$

That question is sufficiently precise to be tested with modern bilateral trade, event, conflict, network, econometric, and machine-learning methods.

References

- [1] Ghosh, S. (2026). *The Financial Flux of a Time-Dependent Interest Rate*. Kolkata, India.
- [2] Ghosh, S. (2026). *The Financial Flow of a Time-Dependent Interest Rate*. Kolkata, India.
- [3] Ghosh, S. (2026). *The Fundamental Identity of a Time-Dependent Interest Rate*. Kolkata, India.
- [4] Ghosh, S. (2026). *The Complete Treatise on the World Economy as a Collection of Sources, Stocks, Flows and Sinks*. Kolkata, India.
- [5] Gaulier, G. and Zignago, S. (2010). BACI: International Trade Database at the Product-Level. CEPII Working Paper No. 2010-23.
- [6] CEPII (2026). *BACI International Trade Database, 202601 Release*. CEPII. https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37
- [7] World Footwear (2025). *World Footwear Yearbook 2025*. APICCAPS, Portuguese Footwear Association.
- [8] World Footwear (2025). Global Footwear Exports Increase in 2024 and China Continues to Lose Share. World Footwear. <https://www.worldfootwear.com/news/global-footwear-exports-increase-in-2024-and-china-continues-to-lose-share/10837.html>
- [9] World Footwear (2026). *World Footwear Yearbook 2026*. APICCAPS, Portuguese Footwear Association. <https://www.worldfootwear.com/news/246-billion-pairs-global-footwear-production-in-2025/11690.html>
- [10] Leetaru, K. and Schrodtt, P. A. (2013). GDELT: Global Data on Events, Location, and Tone, 1979–2012. Paper presented at the International Studies Association Annual Convention.
- [11] GDELT Project (2026). *GDELT Event Database Documentation*. <https://gdeltproject.org/data.html>
- [12] Davies, S., Pettersson, T. and Öberg, M. (2026). Organized violence 1989–2025, and violent political protests. *Journal of Peace Research*, 63(4).
- [13] Uppsala Conflict Data Program (2026). *UCDP Dyadic Dataset, Version 26.1*. <https://ucdp.uu.se/downloads/>
- [14] Feng, L., Huang, Q., Li, Z. and Meissner, C. M. (2026). The “Peace Dividend” of International Trade: A New Empirical Approach. NBER Working Paper No. 35078. <https://www.nber.org/papers/w35078>
- [15] United States Department of State, Office of the Historian (1960). Foreign Relations of the United States, 1958–1960, Volume II: The United Nations and the Republic of the Congo, document concerning the October 1960 General Assembly proceedings. <https://history.state.gov/historicaldocuments/frus1958-60v02/d227>
- [16] United Nations (2023). Did Khrushchev Bang His Shoe at the UN? United Nations Audiovisual Library. <https://www.un.org/ht/video/did-khrushchev-bang-his-shoe-un>
- [17] The Holy Bible. Exodus 3:5.
- [18] The Qur’an. Surah Ta-Ha, 20:12.
- [19] Granovetter, M. (1973). The Strength of Weak Ties. *American Journal of Sociology*, 78(6), 1360–1380.
- [20] Jackson, M. O. (2008). *Social and Economic Networks*. Princeton University Press.
- [21] Newman, M. (2018). *Networks*. Oxford University Press.
- [22] Krugman, P. (1991). *Geography and Trade*. MIT Press.

- [23] Anderson, J. E. (1979). A Theoretical Foundation for the Gravity Equation. *American Economic Review*, 69(1), 106–116.
- [24] Baldwin, R. (2016). *The Great Convergence*. Harvard University Press.
- [25] Keohane, R. O. (1984). *After Hegemony*. Princeton University Press.
- [26] Hirschman, A. O. (1945). *National Power and the Structure of Foreign Trade*. University of California Press.
- [27] Fearon, J. D. (1995). Rationalist Explanations for War. *International Organization*, 49(3), 379–414.
- [28] Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- [29] Lazarus, R. S. and Folkman, S. (1984). *Stress, Appraisal, and Coping*. Springer.
- [30] Frijda, N. H. (1986). *The Emotions*. Cambridge University Press.
- [31] Engle, R. F. and Granger, C. W. J. (1987). Co-integration and Error Correction: Representation, Estimation, and Testing. *Econometrica*, 55(2), 251–276.
- [32] Granger, C. W. J. (1969). Investigating Causal Relations by Econometric Models and Cross-Spectral Methods. *Econometrica*, 37(3), 424–438.
- [33] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
- [34] Newman, M. E. J. (2010). *Networks: An Introduction*. Oxford University Press.
- [35] Barabási, A.-L. and Albert, R. (1999). Emergence of Scaling in Random Networks. *Science*, 286(5439), 509–512.
- [36] Arthur, W. B. (1994). *Increasing Returns and Path Dependence in the Economy*. University of Michigan Press.
- [37] Georgescu-Roegen, N. (1971). *The Entropy Law and the Economic Process*. Harvard University Press.
- [38] Prigogine, I. and Stengers, I. (1984). *Order Out of Chaos*. Bantam Books.
- [39] von Bertalanffy, L. (1968). *General System Theory*. George Braziller.

Glossary

Adaptive Capacity The capacity of an economic relationship or network to absorb a disturbance without entering persistent instability.

Bilateral Dependence The share of an importer’s footwear supply attributable to a particular exporting country.

Diplomatic Stress A measurable deterioration in the quality or character of bilateral diplomatic interactions.

Footwear Flux The temporal derivative of bilateral footwear trade.

Footwear Geopolitical Exposure A measure combining footwear dependence, network concentration, and footwear stress.

Footwear Geopolitical Stability Condition The proposed condition under which adaptive capacity is at least as large as normalized footwear-flow deterioration.

Footwear Stress The negative normalized rate of change of bilateral footwear trade.

Network Centrality A measure of the structural importance of a node in an economic network.

Network Concentration The degree to which trade depends upon a small number of suppliers or buyers.

Network Resilience The ability of an economic network to absorb and recover from perturbations.

Normalized Flux The logarithmic rate of change of a flow.

Symbolic Rupture A visible political act that communicates severe hostility, rejection, humiliation, or institutional defiance.

Shoe Incident A documented political or diplomatic incident in which footwear becomes a prominent symbolic instrument.

Systemic Stress Stress generated by interactions among multiple components of an economic network rather than by a single isolated relationship.

The End